

PROJET VLAN

Packet Tracer

Cabinet Comptafinance

Niveau débutant | 2 VLANs | DHCP | Imprimante partagée

Durée : 6 à 7 heures | Cisco Packet Tracer

Contexte & Objectif

CONTEXTE

Le cabinet **Comptafinance** démarre son informatique et souhaite connecter ses deux services :

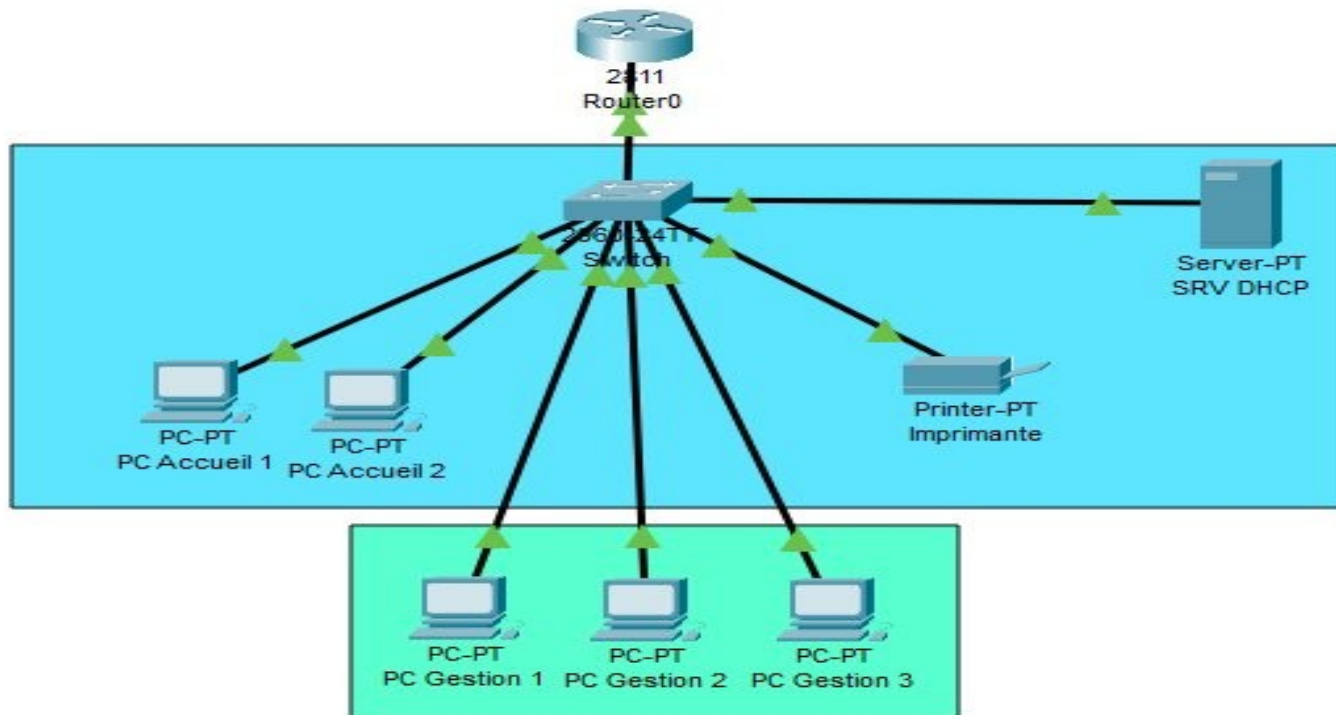
- **Service Accueil**
- **Service Gestion**

Les deux services doivent être isolés mais partager une imprimante et un accès internet.

OBJECTIFS

- ✓ Créer 2 VLANs séparés
- ✓ Configurer un serveur DHCP
- ✓ Partager une imprimante
- ✓ Simuler un accès internet
- ✓ Tester la connectivité
- ✓ Produire les livrables

Schéma réseau — Cabinet Comptafinance



Plan d'adressage IP

Équipement	VLAN	Adresse IP	Rôle
Routeur (fa0/0.10)	VLAN 10	192.168.10.1	Passerelle Accueil
Routeur (fa0/0.20)	VLAN 20	192.168.20.1	Passerelle Gestion
Serveur DHCP	VLAN 10	192.168.10.5	Distribution IPs auto
PC Accueil 1 & 2	VLAN 10	192.168.10.10-50	Postes utilisateurs
PC Gestion 1, 2, 3	VLAN 20	192.168.20.10-50	Postes utilisateurs
Imprimante	VLAN 10	192.168.10.20	Imprimante partagée

Configuration — Switch

Commandes clés :

```
vlan 10
  name Accueil
vlan 20
  name Gestion
interface fa0/2
  switchport mode access
  switchport access vlan 10
interface fa0/24
  switchport mode trunk
```

Switch#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	Accueil	active	Fa0/1, Fa0/2, Fa0/3
20	Gestion	active	Fa0/10, Fa0/11, Fa0/12
1002	fdi-default	active	
1003	token-ring-default	active	
1004	fdi-net-default	active	
1005	trnet-default	active	

Switch#

Résultat : show vlan brief

VLAN 10 Accueil et VLAN 20 Gestion actifs ✓

Serveur DHCP — Attribution automatique des IPs

Pool VLAN 10 — Accueil

Start IP : 192.168.10.10
Gateway : 192.168.10.1
DNS : 8.8.8.8 | Max : 50

Pool VLAN 20 — Gestion

Start IP : 192.168.20.10
Gateway : 192.168.20.1
DNS : 8.8.8.8 | Max : 50

The screenshot shows the configuration interface for 'PC Accueil 1'. It is divided into three main sections: IP Configuration, IPv6 Configuration, and 802.1X. In the IP Configuration section, the 'DHCP' radio button is selected, and a message 'DHCP request successful.' is displayed. The fields for IPv4 Address, Subnet Mask, Default Gateway, and DNS Server are populated with values. The IPv6 Configuration section shows 'Automatic' and 'Static' radio buttons, with 'Static' selected. The 802.1X section has 'Use 802.1X Security' unchecked, and 'Authentication' set to 'MD5'. A 'Top' button is at the bottom left.

IP Configuration	
☒ DHCP	☐ Static
IPv4 Address	192.168.10.2
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

DHCP request successful.

IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::2D0:D3FF:FEAC:1647
Default Gateway	
DNS Server	

802.1X	
☐ Use 802.1X Security	
Authentication	MD5
Username	

☐ Top

DHCP request successful ✓

PC Accueil 1 : 192.168.10.11 — Gateway : 192.168.10.1

Tests de connectivité — Résultats

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping inter-VLAN (Accueil → Gestion)

4/4 paquets reçus — 0% perte ✓

```
C:\>ping 192.168.10.20

Pinging 192.168.10.20 with 32 bytes of data:

Reply from 192.168.10.20: bytes=32 time<1ms TTL=127
Reply from 192.168.10.20: bytes=32 time<1ms TTL=127
Reply from 192.168.10.20: bytes=32 time<1ms TTL=127
Reply from 192.168.10.20: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Ping vers l'imprimante (Gestion → 10.20)

4/4 paquets reçus — 0% perte ✓

Projet terminé avec succès !

Cabinet Comptafinance — Réseau VLAN opérationnel

- ✓ 2 VLANs configurés et isolés
- ✓ DHCP distribue les IPs automatiquement
- ✓ Communication inter-VLAN via le routeur
- ✓ Imprimante accessible depuis les 2 services